Adepu Shiva Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.79
Narayana College	Class XII (Telangana Board of Intermediate Education)	2024	96.80%
Narayana High School	Class X (Board of Secondary Education Telangana)	2022	10.00%

SKILLS

Programming Languages: C, C++, Java, Python, SQL

Technologies: Linux, PostgreSQL, RISC-V, Git, Maven

RELEVANT COURSES

Core CS: Software Development Fundamentals, Data Structures, Algorithms, Database Management Systems

Systems & Theory: Operating Systems, Computer Architecture, Compiler Design, Theory of Computation

Mathematics: Discrete Mathematics for Computer Science, Probability & Random Variables, Linear Algebra

PROJECTS

AI-Driven Contextual Task & Notification Engine (*Flutter, Python, FastAPI, Groq/Llama 3, SQLite, Android SDK*)

- Developed a native Android notification interceptor using **Flutter** and **NotificationListenerService** to capture and parse real-time messaging payloads across **3+ major platforms** (WhatsApp, Discord, SMS).
- Built an asynchronous **FastAPI** backend leveraging **Groq (Llama 3)** with a **5-message sliding context window**, achieving **<200ms processing latency** for multi-turn event extraction in unstructured group chats.
- Designed a **ruthless context-filtering protocol** that successfully filtered out **>85% of casual conversational noise**, isolating high-priority academic deadlines and corporate assessments with **~95% classification precision**.
- Optimized client-side state management using **SQLite** with **semantic context-hash indexing**, eliminating **100% of state duplication glitches** and enabling deterministic, sub-millisecond swipe-to-delete operations.

SentinelAI: Enterprise LLM QA & Observability Platform (Python, FastAPI, PostgreSQL, OpenAI/Anthropic APIs)

- Architected an asynchronous FastAPI backend and PostgreSQL persistence layer for real-time FinTech transcript evaluation, achieving sub-1.5s latency per multi-turn conversation using asyncio.gather for concurrent metric execution.
- Engineered a declarative YAML-based Policy Engine coupled with a hybrid evaluation loop, seamlessly executing deterministic financial boundary checks (Regex/Numeric constraints) alongside probabilistic LLM-as-a-judge metrics.
- Built a Multi-Judge Consensus module leveraging GPT-4o and Claude 3.5 Sonnet to benchmark prompt outputs, tracking inter-judge agreement to isolate model hallucinations, PII leaks, and unsafe policy violations with high confidence.
- Developed a programmatic Synthetic Edge-Case Generator utilizing boundary-value analysis to autonomously bombard LLM pipelines with adversarial scenarios (e.g., jailbreaks, impossible financial states), tracking prompt regression and drift.
- Optimized data persistence and observability using SQLAlchemy 2.0 and JSONB indexing to store complex conversational graphs, enabling granular tracking of token utilization, multi-model inference costs, and execution traces.

xv6 Operating System Enhancements

- Extended the **xv6** operating system across four programming assignments, implementing kernel-level features in process scheduling, virtual memory, and storage systems.
- Developed an **MLFQ scheduler** with multiple priority levels/time slices and added custom system calls.
- Implemented **clock-based page replacement** with disk swapping, encoding swap metadata in page tables and integrating priority-aware eviction.
- Built a **software RAID (0/1/5)** layer with XOR-based parity recovery and implemented **FCFS/SSTF** disk scheduling in the VirtIO driver.

RISC-V Pipelined Processor Simulator (*C++, Computer Architecture*)

- Extended a single-cycle **RISC-V simulator** into a fully functional **5-stage pipelined processor (IF/ID/EX/MEM/WB)**.
- Implemented **hazard detection**, stalling, forwarding, and **static/1-bit dynamic branch prediction** using a Branch History Table (BHT).
- Added a runtime-configurable **L1 data cache** supporting N-way associativity with **LRU, FIFO, and Random** replacement policies.
- Verified correctness with **100% bit-exact GPR/FPR matching** against the single-cycle baseline across **30+ benchmark programs**, achieving **~1.0 CPI** and **87.6% branch prediction accuracy**.

PostgreSQL Query Optimization (*PostgreSQL, Python*)

- Analyzed PostgreSQL query execution plan transitions using the TPC-H benchmark under varying data selectivities.
- Automated execution plan analysis with **Python (psycopg2)** and evaluated optimizer behavior by comparing forced and selected execution plans.
- Used **pg_hint_plan** to force alternative execution plans and evaluate PostgreSQL optimizer decisions using warm-cache execution times.
- Studied execution plan transitions involving **Index Scan, Bitmap Scan, and Parallel Sequential Scan** across multiple benchmark queries.